

COURSE CODE: 23Z013 · ASSIGNMENT PRESENTATION

UNIVERSITY GRANT MANAGEMENT SYSTEM

Blockchain Technology · Subha Subbiah

University grants need auditable state changes

Grant administration can be centralized, or it can make state transitions explicit and auditable.

CENTRALIZED MODEL

One institution controls

One institution controls the database, approvals, and audit trail.

SINGLE POINT OF CONTROL

BLOCKCHAIN MODEL

Smart contract records

The smart contract records grant state transitions and handles allocation and milestone payments.

SHARED, TAMPER-RESISTANT RECORD

Design choice: transparent state changes, programmable rules, and an auditable transaction history—not because centralized systems are impossible to secure.

The contract turns grant administration into a controlled workflow

UNIVERSITY

Creates and funds a grant.

Approves an application.

Adds milestones and
allocates ETH.

Approves; ETH is released.

SMART CONTRACT

Controlled workflow

Grant state, applications, milestones,
allocation, and remaining funds update
through explicit operations.

RESEARCHER

Applies with a proposal.

Submits milestone work.

MetaMask

→ Smart Contract →

Sepolia Blockchain

When remaining funds reach zero, the grant becomes completed.

Three records, one auditable workflow

Grant

grantId · title · totalFunds ·
remainingFunds · researcher

OPEN → ACTIVE →
COMPLETED

Application

application index · applicant ·
proposal

PENDING → APPROVED

Milestone

milestone index · description ·
amount

PENDING → SUBMITTED →
PAID

State transitions are validated on-chain; records preserve funding, selection, and delivery status.

Functions map the roles in the system

Write access follows the actor.
Read-only access exposes the contract state.

ROLE → FUNCTION

UNIVERSITY

Write operations

createGrant()
approveApplication()
addMilestone()
approveMilestone()

RESEARCHER

Write operations

applyForGrant()
submitMilestone()

READ STATE · READ-ONLY ACCESS

getGrant() · getApplicationCount() · getMilestoneCount() ·
getUniversity() · getContractBalance()

Grant functions manage funding

UNIVERSITY

createGrant()

Creates a grant and deposits its funding.

UNIVERSITY

approveApplication()

Assigns the researcher and activates the grant.

ANY READER

getGrant()

Reads grant details.

The university controls funding and researcher assignment; reads expose current grant state.

Application functions handle proposals

RESEARCHER

applyForGrant()

Submits a proposal to a grant.

ANY READER

getApplication()

Reads an application.

ANY READER

getApplicationCount()

Counts applications.

Submission creates a proposal record for university review.

Milestone functions release payments

UNIVERSITY

addMilestone()

Allocates funds within the grant balance.

RESEARCHER

submitMilestone()

Submits work for review.

UNIVERSITY

approveMilestone()

Pays the researcher and updates remaining funds.

ANY READER

getMilestone() · getMilestoneCount()

Reads milestone details and counts milestones.

Supporting reads: `getUniversity()` reads the university address; `getContractBalance()` reads contract balance; `grantCount` tracks grant count.

Tools connect people to on-chain state

Remix IDE

Write, compile, deploy, and call.

MetaMask

University or researcher wallet signs transactions.

Sepolia

Test blockchain holds the contract and test ETH.

Smart contract

Enforces the grant workflow.

Flow: Remix deploys to Sepolia → MetaMask signs university/researcher actions → contract validates and updates state → Remix or frontend reads the result.

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THANK YOU

Blockchain Technology · Subha Subbiah